

FUNDAMENTALS OF NANOELECTRONICS

Basic Concepts

The New Perspective

2. Energy Band Model

3. What and Where

is the Voltage?

4. Heat & Electricity:

Second Law & Information



1.1. Introduction

1.2. Two Key Concepts

1.3. Why Electrons Flow

1.4. Conductance Formula

1.5. Ballistic(B) Conductance

1.6. Diffusive(D) Conductance

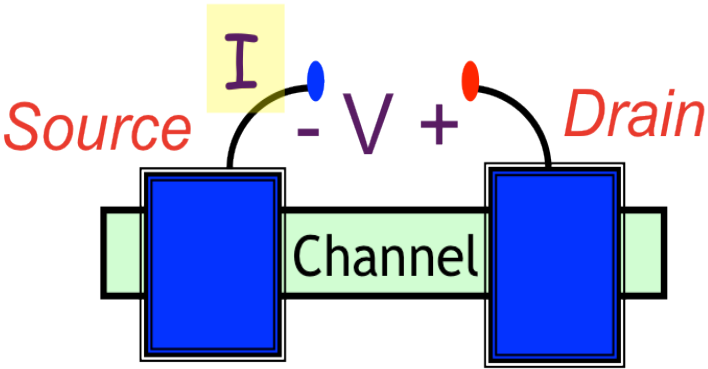
1.7. Connecting B to D

1.8. Angular Averaging

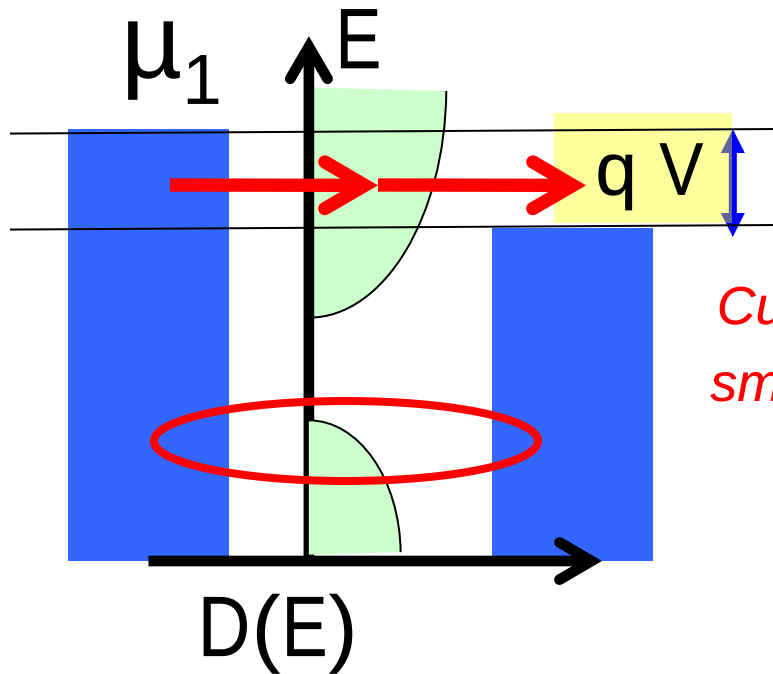
1.9. Drude Formula

1.10. Summing up ..

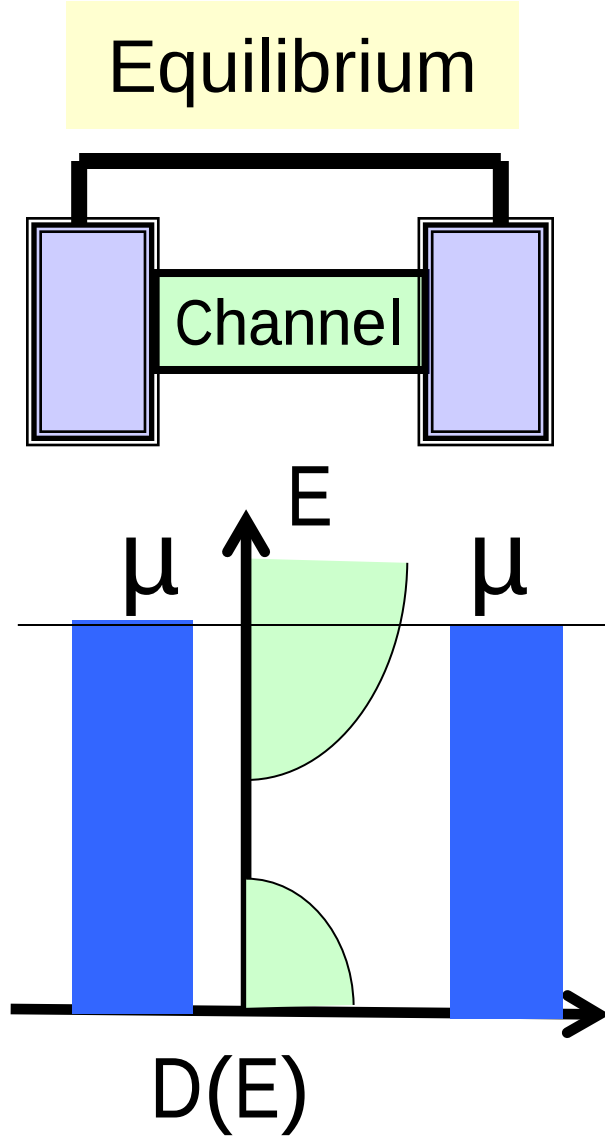
1.3a Why Electrons Flow



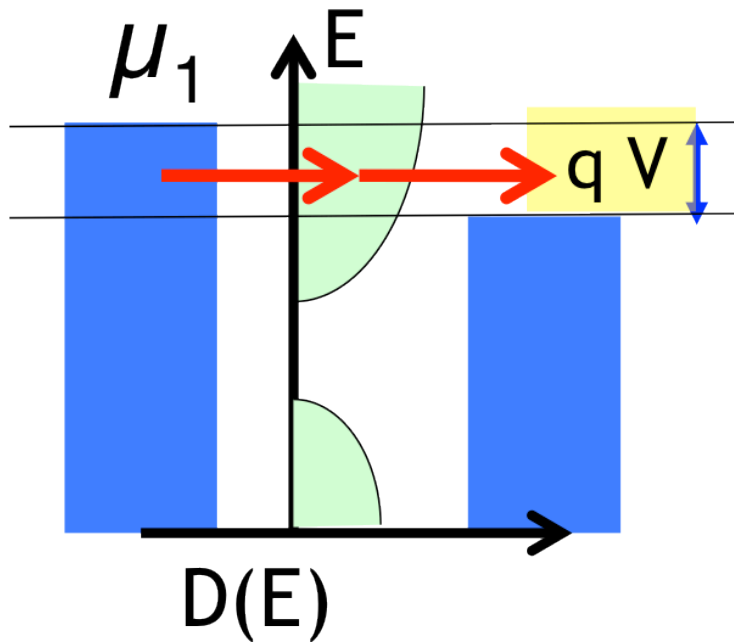
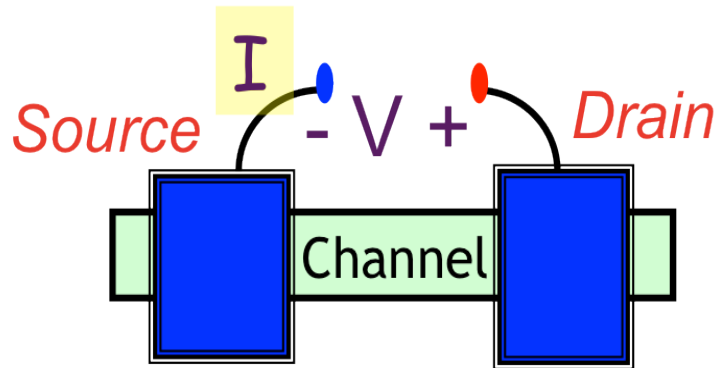
*Hard to understand
if we say that
electric field
drives electrons*



*Current flows only in
small energy window*



1.3b Why Electrons Flow



No energy loss in channel

$$\begin{array}{lll} \text{\# of PhD students} & = & \text{PhD's /year} \quad \text{Time spent getting PhD} \\ \text{\# of electrons in channel} & = & \text{Electrons /second} \quad \text{Time spent in channel} \end{array}$$

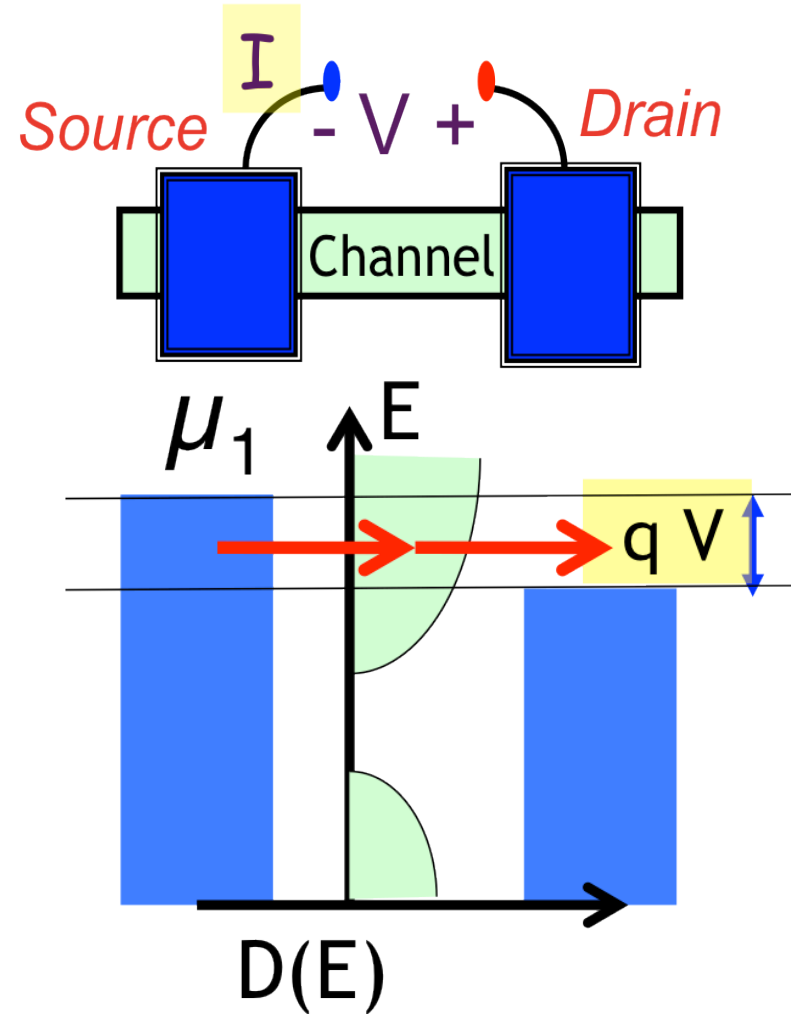
$$\frac{D \cdot qV}{2} = \frac{I}{q} \times t$$

$$G \equiv \frac{I}{V} = \frac{q^2 D}{2t}$$

D: Density of states

t : transfer time

1.3c Why Electrons Flow



Electrons /second $\times$ Time spent in channel = # of electrons in channel

$$\frac{I}{q} \times t = \frac{D \cdot qV}{2}$$

$$\frac{I}{q} \times t(E) = \frac{D(E) dE}{2}$$

$$I = \int_{-\infty}^{+\infty} dE \frac{q D(E)}{2 t(E)} \times (f_1(E) - f_2(E))$$

$$= \frac{1}{q} \int_{-\infty}^{+\infty} dE G(E) (f_1(E) - f_2(E))$$

$$G(E) \equiv \frac{q^2 D(E)}{2 t(E)}$$

$$I = \frac{1}{q} \int_{-\infty}^{+\infty} dE G(E) (f_1(E) - f_2(E))$$

$$G(E) \equiv \frac{q^2 D(E)}{2 t(E)}$$

$$I/V = \int_{-\infty}^{+\infty} dE \left(-\frac{\partial f_0}{\partial E} \right) G(E)$$

Coming up next ..

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